

Technology to Aid Disabled

Using free software, an easy-to-program framework and inexpensive computer tools, Arun Mehta from Delhi has designed a communication system for a child suffering from cerebral palsy condition. The best part is that the technology can be extended and applied for the benefit of many more such children

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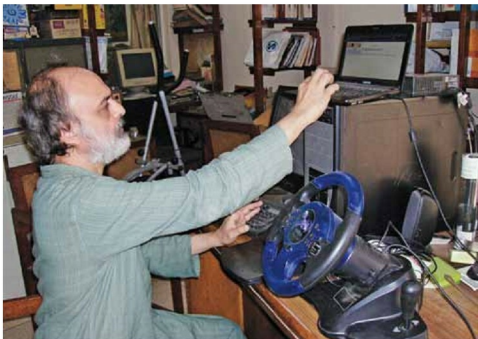
Arpit is a child suffering from cerebral palsy—a condition that restricts him from physical (motor) activities that are quite normal to human beings. He also finds it hard to speak and move.

In today's world, we all are aware of the role that computers have begun to play in our lives. Computers have evolved from being a bulky, stationary machine to handheld devices enabling people work on the move. The dependence on computers has increased so much that being computer illiterate is considered akin to being illiterate altogether. But the situation is very different for people like Arpit, who are suffering from grave disabilities and for whom computing devices offer little help.

It is this challenge that motivated Arun Mehta, a passionate engineer and president of Bapsi, to develop a one-of-a-kind solution to help Arpit learn the basics of communication using computers. In no time, the child was able to type his name, identify pictures, and play games, among other things—just like normal people do.

The technology that drives the 'wheel'

Mehta, an ex-employee of Siemens AG, is a vociferous supporter of open source platforms. He developed the tool, christened 'Arpit's Wheel,' using simple computing hardware and FOSS (free and open-source software) tools and frameworks, which later won him the *Manthan Award* in 2008 in e-inclusion category.



Mehta with 'Arpit's Wheel' tool for the disabled

At first glance, it is hard to believe that the tool is meant to aid a serious case of cerebral palsy. The set-up includes nothing but normal computer hardware, including a PC monitor, CPU, speakers, etc. However, for an input device, Mehta has replaced the traditional keyboard and mouse with a gaming console that we normally see kids playing racing games with. "This is because Arpit cannot control his hand movements, which makes it extremely difficult for him to type or navigate using a mouse," explains Mehta.

Arpit's Wheel runs on a software developed by Mehta. The software is called Skid, and it, along with the framework which it is built on (Ruby on Rails), interacts with the Internet

browser (a version of Firefox). Through a secure Java shell (JSSH), inputs from the game console act as clicks on the web page being displayed in Firefox.

But what is Skid?

The Skid software is mainly a combination of many smaller modules that help the child communicate using computers. The basic modules start by helping the child do elementary tasks such as identify pictures, pick the odd one out, spell words and play educational games. As soon as the child becomes comfortable with the software and different modules, the existing modules could be redesigned or new modules created for advanced communication such as email, instant messaging and Web chat.